Research Method of Data Science

Week 10: Reproducible Preparation and Descriptive Research

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Week 10: Reproducible preparation and descriptive research

Tuesday: how preparation choices shape descriptive evidence

Thursday: a transparent raw-to-analysis pipeline

Real World Bank API data, followed from raw responses to tables and figures.

Companion: **W10_RMDS_Preparation_and_EDA.ipynb**

Agenda

Tuesday — Week 10, Meeting 1

1. A reproducible preparation pipeline
2. Tidy data and safe joins
3. Missing values and analysis samples
4. Descriptive summaries
5. Visual evidence, from simple to richer plots
6. Preparation for Thursday

Appendix: software, function guide, and references

From source validation to an analysis table

Week 9: **Where did the data come from, and can we use them?**

Week 10: **How do our preparation choices affect the findings?**

Pair discussion — 3 minutes

Two researchers download the same source. One fills blanks with zero; the other omits those observations. Could they report different averages? Why?

Pair discussion: numeric example

The same 10 observations can produce different summaries.

Country	GDP	Life	Internet
A	1,000	75	25
B	2,000	85	35
C	NA	95	45
D	4,000	NA	55
E	5,000	115	NA
F	6,000	125	75
G	7,000	135	85
H	8,000	145	95
I	9,000	155	105
J	10,000	165	115

Option 1 · Complete cases

```
df.dropna().mean(numeric_only=True)
```

n = 7
GDP: 6,142.9
Life: 126.4
Internet: 76.4

Option 2 · Replace NA with 0

```
df.fillna(0).mean(numeric_only=True)
```

n = 10
GDP: 5,200.0
Life: 109.5
Internet: 63.5

Discuss: Which choice is defensible—and what assumption does each make?

1. A reproducible preparation pipeline

The example question and data

Question: How do income-related output, life expectancy, and internet use differ across selected countries and over time?

27	24	3	648
selected countries	years, 2000–2023	World Bank indicators	country-years

The notebook embeds the retrieved responses, metadata, queries and hashes.

The measures need their units

GDP per capita PPP

Constant 2021 international dollars

Adjusts purchasing-power differences

Life expectancy

Years

Period's prevailing mortality patterns

Internet use

Percentage of people

Under the provider definition

> GDP per capita does not directly measure household income. Country indicators do not identify individual outcomes.

A reproducible pipeline records decisions

Raw
Preserve API responses and metadata

→

Prepared
Parse, reshape, join, derive

→

Analysis
Define each usable sample

→

Outputs
Tables, figures, decision log

Reproducibility test: the same source snapshot and code should reproduce the same prepared table.

Pipeline challenge — pairs, 3 minutes

3 MIN PAIRS

A student manually edits a downloaded spreadsheet, deletes unusual values, and saves over the original.

Identify two decisions that another researcher could not reconstruct.

Suggest one change that would make the workflow reproducible.

> **OUTPUT:** two unrecoverable decisions and one reproducible alternative

2. Tidy data and safe joins

The observation unit determines the key

A **key** identifies a record at the intended level.

Raw long table

country | year | indicator

Country-year analysis table

country | year

Country-name lookup

country

A country repeated in different years is expected. Repeated full keys need investigation.

Two useful table shapes

Long

country	year	indicator	value
KOR	2000	Internet	44.7
KOR	2023	Internet	97.416

One country–year–indicator measurement per row

Country-year wide

country	year	GDP	life	internet
KOR	2000	…	…	44.7
KOR	2023	…	…	97.416

Separate measurement columns for each country-year

Both structures can be tidy for their intended task.

A join must match its expected cardinality

One ↔ One



GDP country-years + life-expectancy country-years

`one_to_one`

Many → One



Many country-years + one country name per code

`many_to_one`

Use the full key and inspect row counts and unmatched records. A matched key can still contain a missing measurement.

A real join mistake — pairs, 4 minutes

4 MIN PAIRS

Wrong key	Expected key
24 × 24 = ?	24 rows
`country_code` only	`country_code + year`

Korea has 24 GDP records and 24 life-expectancy records, one per year.

The join uses `country_code` alone.

1. How many rows result?
2. What comparison does each row now represent?
3. What key and validation should replace it?

OUTPUT: row count, interpretation, and corrected join key

A short join that documents expectations

```
joined = gdp.merge(
    life,
    on=["country_code", "year"],
    how="outer",
    validate="one_to_one",
    indicator=True
)
```

Check `_merge`, unique keys, row count, and missing values after the join.

3. Missing values and analysis samples

The snapshot contains genuine missing values

Across **648 country-years**:

48	0	26
GDP per capita PPP missing measurements	Life expectancy missing measurements	Internet use missing measurements

The expected country-year keys are present. Some measurements are unavailable.

Missing-data decisions — pairs, 4 minutes

4 MIN PAIRS

Your figure compares GDP and life expectancy in 2023. Internet-use data are also in the table.

1. Which variables must be observed for this figure?
2. Should a missing internet value exclude a country?
3. What could go wrong with replacing every missing measurement with a global mean?

> **OUTPUT:** required variables, usable sample rule, and one risk of mean filling

The class preparation policy

1	2	3	4	5
Keep expected country-years	Preserve missing values	Report usable N	Document changes	Investigate unusual values

A different policy requires a reason and a visible code change.

The sample changes with the question



A report should name its denominator rather than simply saying “the dataset has 27 observations.”

4. Descriptive summaries

A summary needs a stated sample

Choose the unit and period before calculating:

One country	Cross-section	Pooled panel
over time	countries in one year	repeated country-years

These answer different descriptive questions.

An unweighted country mean gives each observed country equal weight.

Location and spread tell different things

Mean

Arithmetic average of observed values

Median

Middle of the ordered distribution

Standard deviation

Spread around the mean

IQR

75th percentile minus 25th percentile

Read them alongside **N, missing count, minimum, and maximum**.

A real 2023 summary

Measure	Usable N	Mean	Median
GDP per capita PPP	25	34,261	32,858
Life expectancy, years	27	76.00	76.66
Internet use, %	25	82.97	86.42

GDP units: constant 2021 international dollars.

Source: World Bank WDI API snapshot, retrieved 31 October 2026.

Summary interpretation — pairs, 4 minutes

4 MIN PAIRS

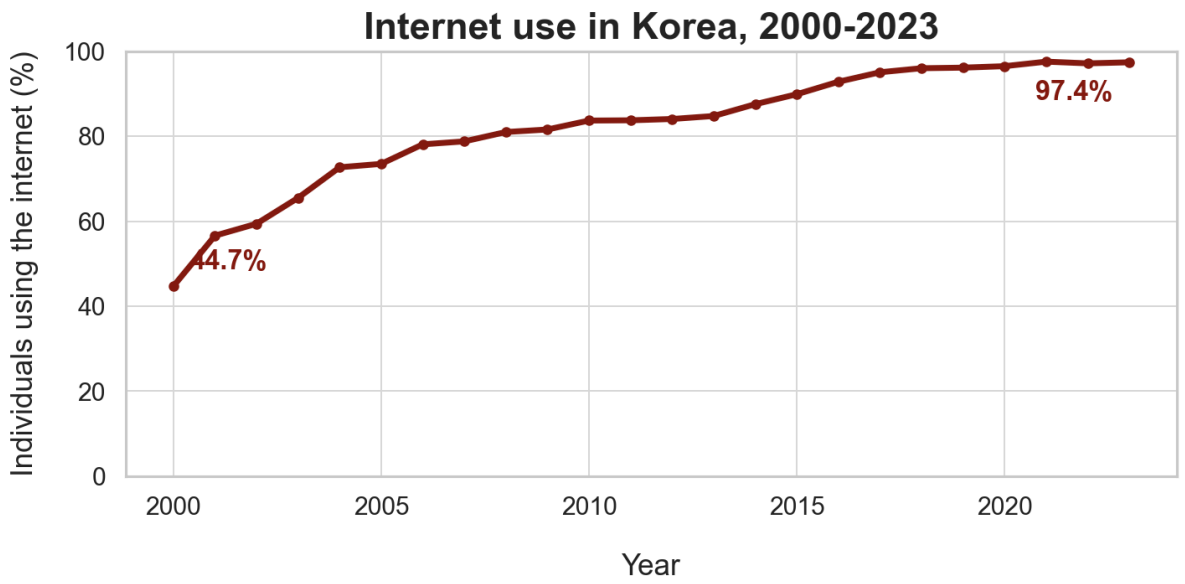
Using the summary:

1. Write one sentence comparing mean and median.
2. Suggest what to inspect in the distribution. , Definition \@ref(def:label)
3. Explain why this is not the percentage of all people worldwide who use the internet.

> **OUTPUT:** one comparison, one inspection target, and one limitation

5. Visual evidence from simple to richer plots

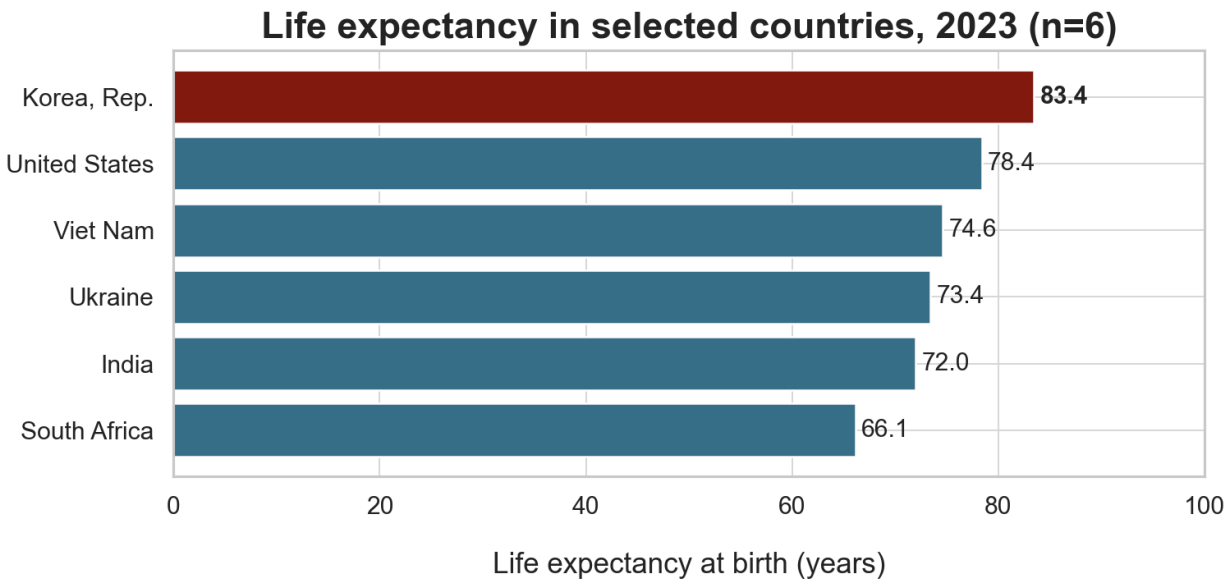
A line shows change over time



Source: World Bank WDI, IT.NET.USER.ZS; retrieved 31 Oct 2026.

READ THE CHART: What changed, over which period, and in which units?

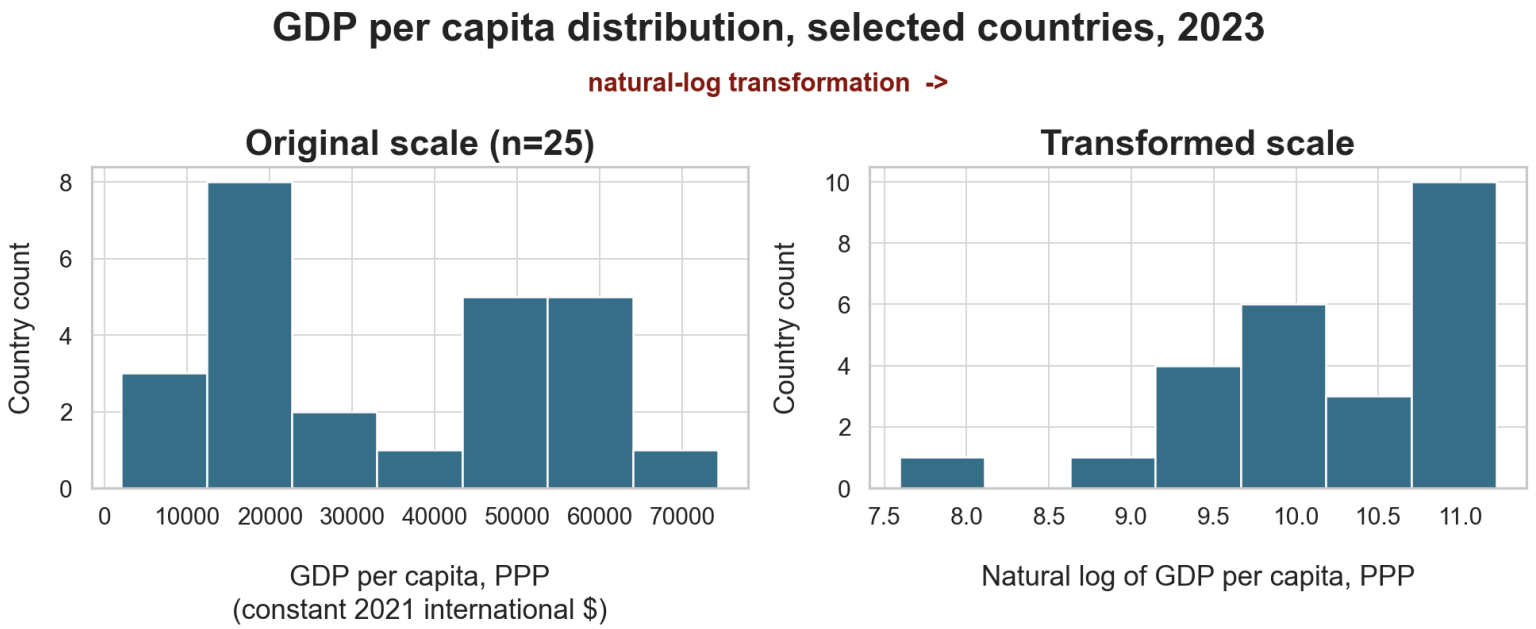
A small comparison can use simple bars



Source: World Bank WDI, SP.DYN.LE00.IN; retrieved 31 Oct 2026.

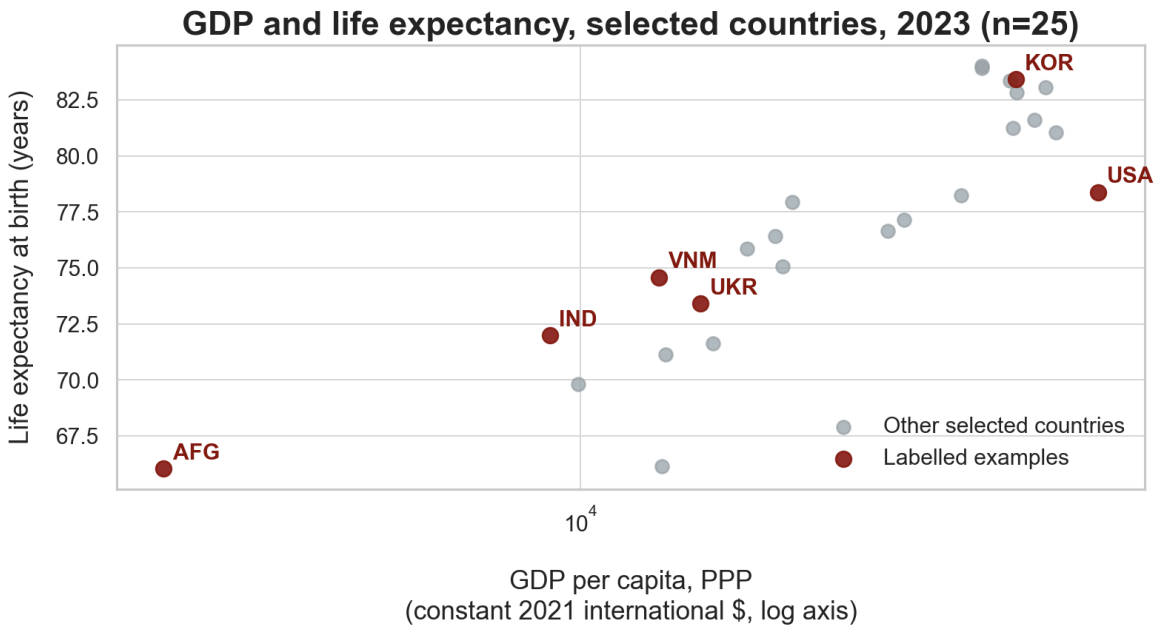
READ THE CHART: Why does the bar axis start at zero? What does the selection omit?

A distribution depends on the scale



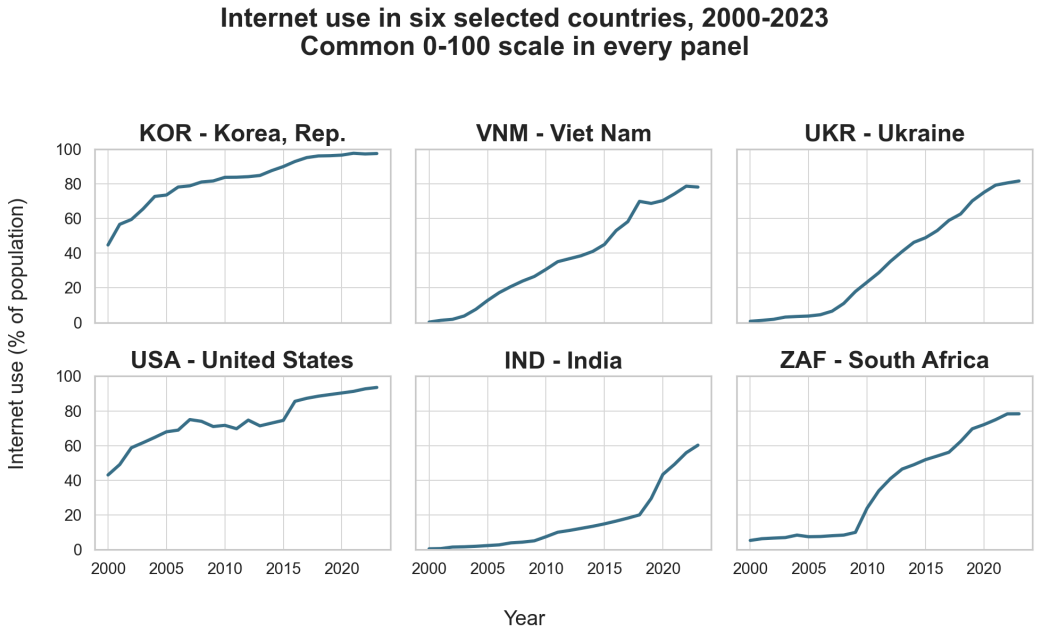
READ THE CHART: Same countries and values. How does the natural-log transformation change the view?

A relationship needs a bounded claim



READ THE CHART: Each point is one country in 2023. A log x-axis changes spacing.

Small multiples reduce line clutter



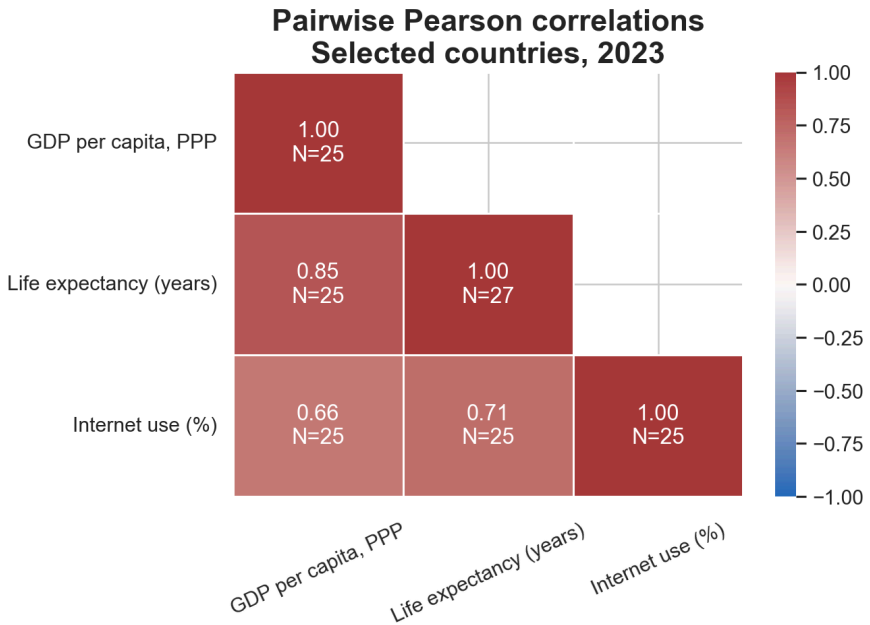
READ THE CHART: Which comparisons are easier with common axes and separate country panels?

Descriptive reports with reusable functions

Function / tool	Output
`df[columns].describe()`	Counts, mean, spread, quartiles, extremes
`numeric_summary()`	Also missingness, IQR, skewness, inspection flags
`descriptive_report()`	Numerical tables, histograms, boxplots, local HTML report
`sns.pairplot()`	Scatterplot matrix and distributions
Optional `ydata-profiling`	Broad automatic local report

Choose measurement columns and the sample explicitly.

A correlation also needs its usable N



READ THE CHART: Does each coefficient use the same variables and observations as the previous plot?

Caption challenge — pairs, 2 minutes

2 MIN PAIRS

Choose one chart from this section.

Prepare a two-sentence caption containing:

- measure, unit, period and sample
- one descriptive pattern and one limitation

Avoid attributing a causal mechanism that the figure cannot establish.

> **OUTPUT:** a two-sentence caption with context, pattern, and limitation

6. Preparation for Thursday

The studio output

Bring your source audit and approved project data, if ready.

The notebook supplies real API records for everyone.

Prepare	Describe	Interpret
Transparent decision log and validated table	Numeric summary and two suitable plots	Short finding with sample and limitations

Use your approved project or a justified subset of the class data.

Thursday – Cleaning and visualization studio

Agenda

Thursday — Week 10, Meeting 2

1	→	2	→	3	→	4	→	5
Setup		Prepare		Describe		Apply		Review

Work individually. Use pairs for checking decisions.

Setup — 5 minutes

STEP 1 OF 7 | 5 MIN

Open **W10_RMDS_Preparation_and_EDA.ipynb**.

Save a personal copy and run Section 0.

Keep `REFRESH_API=False` for the embedded, verified source snapshot.

Expected scope: 27 countries, 24 years, three indicators.

The standard run requires no separate data files or live API call.

> **EXPECTED ARTIFACT:** personal notebook copy with Section 0 completed

Preparation and joins — 15 minutes

STEP 2 OF 7 | 15 MIN

Notebook Sections 1–2

1. Read the indicator definitions and source note.
2. Run the parser and tidy reshape.
3. Compare the wrong and validated joins.
4. Complete Checkpoint A.

Explain why **576 rows** is a warning in the Korea example.

> **EXPECTED ARTIFACT:** validated country-year analysis table and Checkpoint A

Missingness and transformations — 10 minutes

STEP 3 OF 7 | 10 MIN

Notebook Section 3

1. Inspect missing values by country and variable.
2. Compare total rows with usable observations.
3. Read the class missing-data policy.
4. Inspect log GDP and endpoint percentage-point change.
5. Complete Checkpoint B.

EXPECTED ARTIFACT: documented missingness decision and derived measures

First summaries and plots — 12 minutes

STEP 4 OF 7 | 12 MIN

Notebook Sections 4–5

1. Run the numeric summary.
2. Make the Korea line and small country comparison.
3. Run the histograms and scatterplot.
4. Change one visual choice and complete Checkpoint C.

Use the supplied code. Explain the consequence of your change.

> **EXPECTED ARTIFACT:** numeric summary, first plots, and Checkpoint C

A broad descriptive report — 10 minutes

STEP 5 OF 7 | 10 MIN

Notebook Section 6

Run `descriptive_report()` on the 2023 data.

Inspect:

- usable N and missingness
- median, IQR and inspection flags
- histograms and boxplots
- correlations alongside pairwise sample sizes

Open or download the local HTML report.

> **EXPECTED ARTIFACT:** local descriptive report with tables, plots, correlations, and N

Individual project studio — 15 minutes

STEP 6 OF 7 | 15 MIN

Notebook Sections 8–9

1. Run the reproducibility check and inspect the decision log.
2. Define your project unit, key and preparation decision.
3. Produce a summary and two suitable plots.
4. Draft a 100–150 word descriptive finding.

If your own data are unavailable, choose an explained subset of the class API data.

> **EXPECTED ARTIFACT:** project summary, two plots, and a 100–150 word finding

Peer review — 5 minutes

STEP 7 OF 7 | 5 MIN

Two minutes per student, then one minute to revise.

Partner checks:

- row unit and join keys
- missing-data policy and reported sample
- plot labels, scales and units
- whether the finding stays within the evidence

Record one revision or a justified decision to retain your approach.

> **EXPECTED ARTIFACT:** one recorded revision or justified decision

Save and submit — 3 minutes

SUBMIT 3 MIN

Save your completed notebook.

Download the generated output bundle with raw provenance, prepared data, logs, tables and figures.

Include your descriptive finding, peer revision, and AI-use/verification statement.

Follow the posted Cyber Campus instructions.

> **FINAL CHECK:** notebook, output bundle, finding, peer revision, and AI-use statement

Appendix — reference and stretch material

Notebook map

0–1
Setup and API snapshot

→

2
Parse, reshape, join

→

3
Missingness and measures

4–5
Summaries and plots

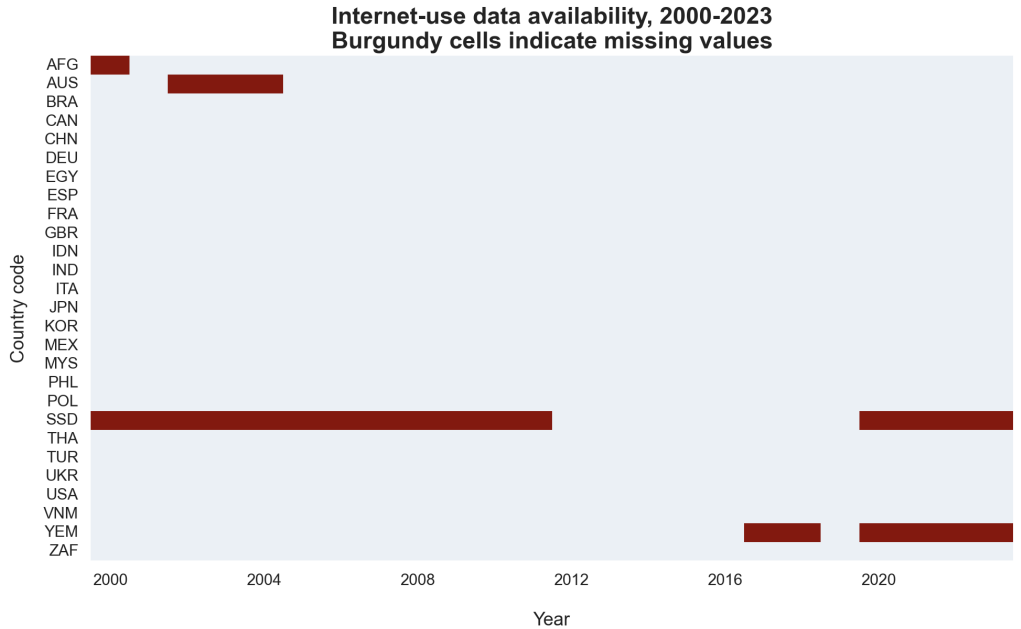
→

6–7
Reports and stretch

→

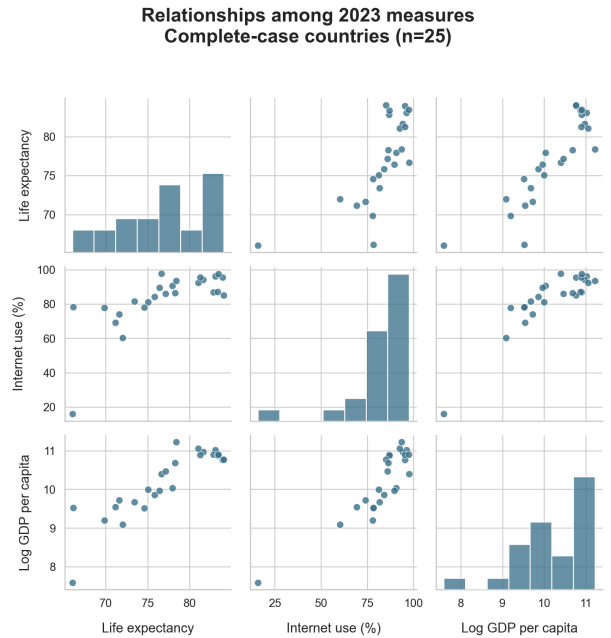
8–10
Rerun, project, submit

Stretch: missingness over countries and time



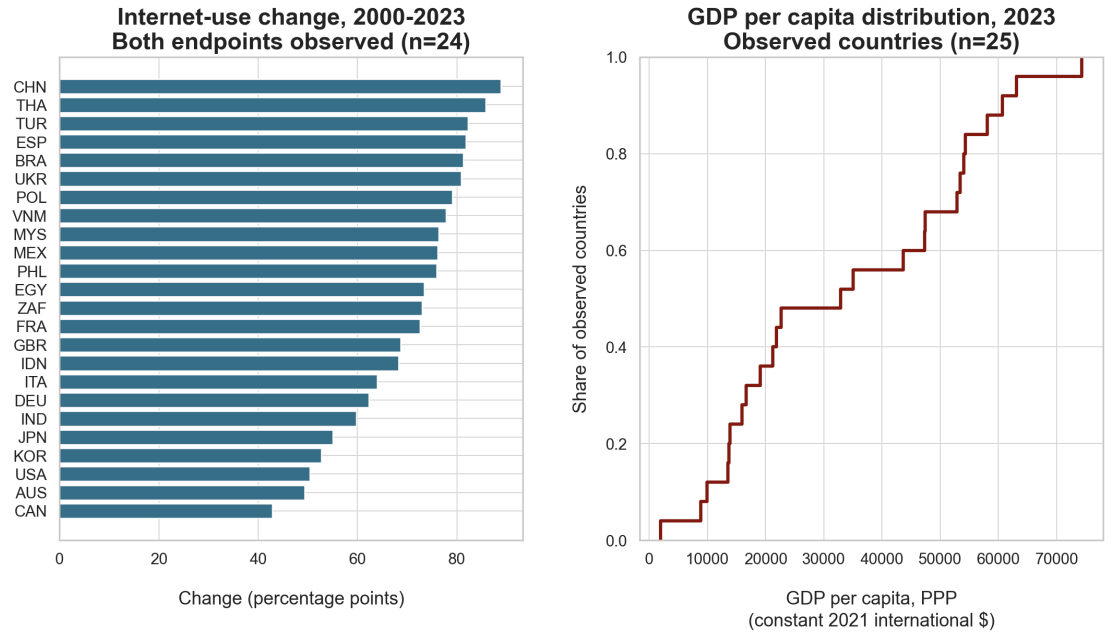
READ THE CHART: Colour encodes missingness, not internet-use level.

Stretch: a scatterplot matrix



READ THE CHART: Same 25 complete-case countries in every panel. GDP uses a natural-log transformation.

Stretch: endpoint change and the ECDF



READ THE CHART: Change requires both endpoints. The ECDF uses observed GDP values in the selected 2023 sample.

What the report functions return

`numeric_summary()`

Rows, usable N, missing count/percentage, unique values, mean, SD, quartiles, tail percentiles, IQR, skewness and inspection flags.

`descriptive_report()`

Summary, schema, missingness, pairwise correlations and N, histograms, boxplots, CSV tables and a local HTML report.

Examples of defensible descriptive language

"Among the **25 selected countries** with observed GDP **in 2023**, the mean GDP per capita PPP was about **34,261 constant 2021 international dollars**."

"The internet-use mean was below its median **in this sample**, prompting inspection of the lower part of the distribution."

State the **sample, period, units and limitation** in the paper.

Published research example

Vyshnevskyi, I., & Sohn, W. (2023).
"Nonperforming loans and related lending: Evidence from Ukraine." *Emerging Markets Review, 57*, 101069.

Open PDF | Publisher page

Scroll inside the paper below. As you read, look for how the authors connect the research question, data preparation, descriptive evidence, and claims.



Software and source references

Data and indicators

- World Bank WDI API
- GDP per capita PPP
- Life expectancy
- Internet use
- World Bank dataset terms

Software documentation

- pandas: describe
- pandas: validated merge
- seaborn: pairplot
- Optional YData Profiling

DATA NOTE World Bank WDI and underlying providers. API snapshot retrieved 31 October 2026.